

School of Computing and Mathematical Sciences

CO7201 Individual Project

Interim Report

An Online Tool to Manage Pension Benefits

Shweta Bhagat

sbb24@student.le.ac.uk

239056945

Project Supervisor: Muhammad Iftikhar
Principal Marker: Dr Martina Palusa

Word Count: 750
28 March 2025

DECLARATION
All sentences or passages quoted in this report, or computer code of any form whatsoever used and/or submitted at any stages, which are taken from other people's work have been specifically acknowledged by clear citation of the source, specifying author, work, date and page(s). Any part of my own written work, or software coding, which is substantially based upon other people's work, is duly accompanied by clear citation of the source, specifying author, work, date and page(s). I understand that failure to do this amounts to plagiarism and will be considered grounds for failure in this module and the degree examination as a whole.

Name: Shweta Bhagat
Date: 28 March 2025

Contents

1. **Overview** 3
2. **Task Progress Status** 4
3. **Updated Timeline** 5

1. Overview

Project Requirements:

The system requirements for "An Online Tool to Manage Pension Benefits" are based on the functionality needed by pensioners in the UK pension system. The key requirements identified in the Preliminary Report are as follows:

Essential Requirements:

- Secure user authentication and role-based access (pensioner/admin)
- Multi-Factor Authentication (MFA)
- View pension statements and contribution history
- Update personal information securely
- Compliance with UK pension regulations and data protection standards (e.g., GDPR)

Desirable Features:

- Real-time notifications and alerts (e.g., email notifications for pension disbursement)
- Analytics dashboard with charts for pension insights and projected retirement value
- Automated pension contribution calculations based on salary and investment type
- Document storage and management for pension-related files

Optional Features:

- Multi-language support for inclusivity
- AI-powered chatbot for FAQs and pension guidance
- Gamification elements to promote financial literacy (e.g., badges or points for completing profile or reading pension tips)

These requirements serve as a foundation for designing and implementing the system and have guided all design and implementation decisions to date.

This interim report provides a project update on the implementation of "An Online Tool to Manage Pension Benefits", a secure and interactive web platform designed to support UK pensioners in managing pension data efficiently. The project, based on the requirements stated in the Preliminary Report, includes secure multi-factor authentication, role-based access, pension tracking, profile management, document handling, and compliance with UK regulatory guidelines.

Progress has been made in core backend development, secure user authentication, and Angular frontend setup. A bold and accessible UI design has been finalized for the homepage and dashboard. The Flask backend has been connected to a MySQL database, and major authentication components have been implemented, including JWT and MFA. Frontend-backend connectivity has been verified, and the homepage UI is currently under construction.

The system remains under development for pension statement tracking, profile editing, admin dashboard integration, and upcoming features like document management, notifications, and analytics. An updated time plan has been provided with revised deadlines based on current progress.

2. Task Progress Status

1. Completed Tasks

- Flask Backend Setup
  - Successfully created app, config.py, wsgi.py, and initial folder structure.
  - Verified project runs correctly in a local environment.
- Database (MySQL) Setup
  - Created and configured MySQL database.
  - Added and tested connection between Flask and MySQL.
- User Authentication System (with MFA & RBAC)
  - Implemented JWT-based login and signup.
  - MFA via QR code using PyOTP.
  - Role-based access control for user and admin.
- Initial Frontend Setup (Angular)
  - Created Angular project with routing.
  - Configured API connectivity with Flask backend.
  - Verified login/signup functionality connects correctly to backend.
- Preliminary Homepage Design (UI planning)
  - Completed full homepage design plan with sections, animations, styles, and theme colours.
  - Finalized styling decisions (no minimalism, bold UI, icons, transitions, animations).

2. Ongoing Tasks (in progress)

- Admin Dashboard Functionality
  - Currently building functionality to add pensioners, assign pension plans, and view users.
  - Backend routes prepared; Angular form pages under development.
- User Dashboard UI Design
  - Designing dashboard with pension contribution graphs, cards, and projections.
  - Integrating Chart.js for animated financial visuals.
- Homepage Angular Implementation
  - Layout and section structure in progress:
    - Hero, Features, Counters, Testimonials completed
    - Remaining: FAQs, Footer, Parallax animations

3. Pending Tasks (yet to start)

- Pension Plan Management System
  - Admin to define pension plans (rules, amounts, disbursement schedule).
  - Backend logic + Angular interface pending.
- Contribution History & Projection Engine
  - Implementation of automated pension calculation based on user salary, tax brackets.
  - Storage and graph generation of monthly contributions.
- Document Management Module
  - Secure upload, storage (e.g. Firebase Storage), and retrieval of pension-related documents.
- Email Notification System
  - Set up backend alerts and reminder system for contribution updates and policy changes.
- Deployment & Final Testing
  - Deploy frontend to Firebase
  - Backend hosting + domain integration
  - Perform full system testing & user acceptance tests

3. Updated Timeline

Phase	Task	Original Duration	Updated Completion Date	Status
Requirement Analysis & Planning	Define system objectives, scope, feasibility	1 week	24 Feb	Completed
	Identify pension management functionalities	1 week	2 Mar	Completed
	Roadmap and risk plan development	1 week	9 Mar	Completed
System Design	Design architecture, workflows, and compliance checklists	1 week	16 Mar	Completed
	Database schema and security framework	1 week	23 Mar	Completed
Implementation (Core Dev)	Authentication with MFA & RBAC	2 weeks	6 Apr	Completed
	Admin backend functionality	1 week	8 Apr	In Progress
	Homepage frontend sections	1 week	10 Apr	In Progress
Feature Enhancement	Pension statement and tracking module	1 week	15 Apr	Pending
	Profile update feature	4 days	17 Apr	Pending
	Analytics dashboard	5 days	20 Apr	Pending

	Automated contribution calculation	4 days	22 Apr	Pending
	Document management	3 days	24 Apr	Pending
	Email alerts and notifications	3 days	25 Apr	Pending
Testing & QA	Functional, security, and UAT testing	3 days	27 Apr	Pending
Final Documentation	Final report and submission	–	30 Apr	Pending